

## EARTHPRESS · DESIGN DECISION RECORD

# EarthPress recommendations accepted

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## DOCUMENT CONTROL

|             |                                                               |         |                |
|-------------|---------------------------------------------------------------|---------|----------------|
| Document ID | EPR-DDR-002                                                   | Version | 0.1            |
| Status      | <b>Draft</b>                                                  | Date    | 2026-09-25     |
| Author      | Amish Chadha                                                  | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                                 |         |                |
| Repository  | github.com/BoujeeEnjinia1701/earthpress                       |         |                |
| Source      | docs/decisions/0002-recommendations-accepted.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                  |
|---------|------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 2026-09-25 | Amish Chadha | Record Amish's acceptance of every recommendation in EPR-DDR-001 and docs/REVIEW.md, what changed in the repo, and the items still open |

# 0002: Recommendations accepted

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- **Date:** 2026-09-25
- **Status:** accepted. Every item with a recommendation is decided; item 8 (first co-design partner and region) has no recommendation and remains proposed, awaiting Amish.

## ■ Context

EPR-DDR-001 adopted the TRL 2 recommendations for TRL 3 work, open for Amish's review, and listed seven new TRL 3 items (10 to 16) as proposed, awaiting Amish. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Each item with a recommendation is therefore **decided by Amish, 2026-09-25: go with recommendation**. Where an item offered several options, the recommended option is the decision. Items with no recommendation stay open. The project stays at TRL 3 ( `trl: 3` , `trl_target: 3` ); TRL 4 work is on hold by Amish's instruction.

## ■ Decision

*Table 1. Items decided on 2026-09-25 and what changed in the repo.*

| #  | Item                  | Decision                                                                                                                                                                                                                                       | What changed                                                                                                                       |
|----|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Mechanism             | Bottom piston, toggle and one lever (CINVA-Ram pattern); bottle-jack variant documented later                                                                                                                                                  | None; already in the precis and model                                                                                              |
| 2  | Lever layout          | Longer arc and higher end grip, met in intent with a low pivot (59.2° arc, grip 1.89 to 0.86 m)                                                                                                                                                | None; already in the model                                                                                                         |
| 3  | Block                 | 290 x 140 x 90 mm; interlocking inserts later                                                                                                                                                                                                  | None                                                                                                                               |
| 4  | Compaction target     | 2 MPa                                                                                                                                                                                                                                          | None                                                                                                                               |
| 5  | Stabilizer            | 5 % cement, lime for clay-rich soils, confirmed with partner soils                                                                                                                                                                             | None; partner soil confirmation is TRL 4 work, on hold                                                                             |
| 6  | Meeting R7            | 12 mm base plate and bolted mold                                                                                                                                                                                                               | None; already in the model (heaviest piece 36.6 kg)                                                                                |
| 7  | Soil test kit         | Field tests only                                                                                                                                                                                                                               | None                                                                                                                               |
| 9  | Pitch and problem     | No rewording                                                                                                                                                                                                                                   | <code>project.yaml</code> pitch and problem unchanged; the budget part is superseded by item 14                                    |
| 10 | Ejection              | (a) Separate eject seesaw, same lever in a second socket, 200 mm lost-motion slot                                                                                                                                                              | None; already in the model                                                                                                         |
| 11 | Two operators         | (a) Two people on a T-handle, 334 N each                                                                                                                                                                                                       | Precis and requirements now describe the press as a two-person machine                                                             |
| 12 | R5 wording            | 500 N or less per operator; grip 0.8 to 1.9 m; peak force at 1.0 m or higher                                                                                                                                                                   | EPR-REQ-001 R5 reworded (was 500 N, grip 0.9 to 1.6 m); status <b>not met to met on paper</b> (334 N per operator, peak at 1.16 m) |
| 13 | R7 total              | (a) Relax total to 190 kg, keep the 50 kg piece limit                                                                                                                                                                                          | EPR-REQ-001 R7 total 140 kg to 190 kg; status <b>not met to met on paper</b> (187 kg, 3 kg margin)                                 |
| 14 | Budget                | (a) Raise <code>budget_usd</code> to \$500                                                                                                                                                                                                     | <code>budget_usd</code> \$450 to \$500; R13 target \$450 to \$500; status <b>not met to met on paper</b> (\$488, \$12 margin)      |
| 15 | Fill control          | Weigh every fill on a 10 kg scale to about +1.3 % ± 100 g; target mass on the test chart                                                                                                                                                       | None; the scale is already in the BOM and the rule in the precis                                                                   |
| 16 | Engineering proposals | (a) Pivot at x = -468, z = 266 mm, 1.7 m lever, rim at 940 mm; (b) crank stop at 6° (173 kN design load); (c) ribbed lid; (d) end pawl; (e) linkage guard; (f) case-hardened bushes; (g) R4 at 296 blocks a day, revisit when cycles are timed | None to geometry; all were already in the model. R4 stays at risk; timing cycles is TRL 4 work, on hold                            |

No decision changed the geometry, so `cad/src/model.py`, the STEP and STL exports and EPR-DWG-001 keep their geometry and notes; the drawing stays at Rev P1 and was regenerated only for the title block. EPR-CAL-001 moves to v0.2: the script now checks the accepted R5 band and peak height, the 190 kg R7 total and reads the budget from `project.yaml`.

## Items still open

| # | Item                                                                                         | Status                                               |
|---|----------------------------------------------------------------------------------------------|------------------------------------------------------|
| 8 | First co-design partner and region (examples: an earth-building NGO in East Africa or India) | Proposed, awaiting Amish; no recommendation was made |

## On hold (TRL 4)

Decided in principle but not started, because TRL 4 is on hold: confirming the stabilizer with partner soils (item 5), timing the press cycle for R4 (item 16g), and every build, block test, proof-load test and measurement of the soil compaction curve.

## ■ Consequences

- EPR-PRB-001, EPR-PRC-001 and EPR-REQ-001 move to v0.4; EPR-CAL-001 to v0.2; EPR-DDR-001 to v0.2.
- Requirements on paper: none not met; R2 and R4 at risk; R3, R8, R10 and R14 not verifiable at TRL 3; R1, R5, R7, R11, R12 and R13 met on paper; R6 and R9 met by design review.
- R7 and R13 are met with small margins (3 kg and \$12). Any growth in mass or price reopens them.
- No cross-repo actions: EarthPress uses no shared component.